

Attribute Value Extraction

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Abstract

Over the past few years, numerous studies have explored product value extraction, with large language models (LLMs) emerging as top performers. A 2020 study evaluated two distinct methods — Infilling by Language Modeling (ILM) using GPT-2 and Question Answering finetuning with T5—highlighting inconsistencies due to differing underlying model and their architectures. More recently, the Extract framework showcased cutting-edge results with GPT-4 and GPT-3.5. However, hosting such models poses significant challenges for many organizations. To address this, we leveraged the lightweight, open-source LLaMA 8B model for a comprehensive and accessible analysis.

1. Introduction

Product attributes and their values play a critical role in e-commerce platforms, as they help define and categorize products in a way that makes them more accessible to customers. With hundreds of thousands of products available online, each product category comes with its own unique set of attributes. These attributes are essential for enabling customers to search for, compare, and ultimately select products that best meet their needs.

Every day, new products are added to e-commerce catalogs, often introducing new attributes or values that were not seen previously. However, product information is frequently incomplete, inconsistent, or noisy, with key attribute details either missing or poorly specified. Given the vast diversity of products and the constant addition of new ones, it becomes increasingly important for attribute extraction methods to adapt to this variability. To address this, such methods must operate under an **open-world assumption**, meaning they should be capable of identifying and handling attributes and values that have not been encountered before.

Previous studies have approached the task of attribute value extraction using different methodologies. One of the earlier works (Roy et al. [1]) formulated the problem as two distinct tasks: *text infilling* and *answer generation*. It utilized separate models for these tasks, specifically GPT-

2 (small) for text infilling and T5 for answer generation. However, this approach was unable to determine which method was more effective, as the use of different models introduced inconsistencies in comparison.

More recent efforts (Brinkmann et al. [2]) have leveraged advanced Large Language Models (LLMs) like GPT-3.5 and GPT-4 to extract attribute values. These works employed carefully designed prompts using strategies such as *zero-shot prompting* and *few-shot prompting*. While these methods have demonstrated success, the reliance on large-scale models makes them computationally expensive.

In contrast, this work investigates the use of a computationally lightweight and open-source model, the **LLaMa 8B**, to efficiently perform attribute value extraction. By employing the same model for both *text infilling* and *answer generation*, we ensure a consistent evaluation of the two approaches. Additionally, we evaluate the robustness of fine-tuned models by testing them on datasets different from their training data, enabling an analysis of their generalization capabilities. The main contributions of this work are as follows:

- Employing open-source and computationally efficient LLMs, such as the LLaMa 8B, for attribute value extraction.
- Conducting a comparative study of the *text infilling* and *answer generation* approaches using the same LLM, ensuring a consistent evaluation.
- Evaluating the generalization ability of fine-tuned models through cross-dataset testing, comparing their performance to that of the base LLMs.

Consider a product description: *"The XYZ Electric Guitar features a solid mahogany body, 6 strings, dual-coil pickups, and a sleek matte black finish. Ideal for beginners and professionals alike."* From this description, the extracted attributes and their corresponding values could be:

- **Body Material:** Mahogany
- **Strings:** 6
- **Pickup Type:** Dual-coil
- **Color:** Matte Black

2. Problem Statement

In this section, we formally define the problem of attribute value generation. Given a product context $T = (w_t^1, w_t^2, \dots, w_t^m)$ and its attribute $A = (w_a^1, w_a^2, \dots, w_a^n)$, the goal is to generate the corresponding value $V = (w_v^1, w_v^2, \dots, w_v^e)$. For example, consider the product context: "Samsung Galaxy S23 Ultra 5G Smartphone 256GB Storage 12GB RAM Quad Camera Snapdragon 8 Gen 2." If we focus on the attributes "Model" and "Battery Capacity," the desired output is the value "Galaxy S23 Ultra" for the attribute "Model" and NULL for the attribute "Battery Capacity," as this information is not included in the context.

In this work, we first formulate this problem as a (i) text infilling task and then as an (ii) answer generation task. In the text infilling approach, we combine the context T , the attribute A , and the value V into a sentence of the form "T. A is V." where the value V is masked. Our goal is to generate the missing span, thereby predicting the value. Let the incomplete sentence be $\tilde{S} = (w_s^1, w_s^2, \dots, w_s^p)$. The model outputs the best attribute value sequence $\tilde{V}$ by learning the distribution $\tilde{V} = P(V|\tilde{S})$. In the answer generation approach, the objective is to generate V as the answer, with T as the context and A as the question.

3. Experimental Setup

3.1. Dataset

We utilize two benchmark datasets, OA-Mine and AE-110K, consisting of English product offers annotated with attribute-value pairs. These datasets are widely used for evaluating tasks related to attribute-value extraction. The MAVE dataset is excluded from consideration, as its attribute-value pairs were annotated using an ensemble of fine-tuned models and lack manual verification. Upon inspection, numerous annotation errors were found in the AE-110K dataset, whereas the OA-Mine dataset, being human-annotated, did not exhibit such issues.

OA-Mine. The OA-Mine (Zhang et al. [3]) dataset contains human-annotated product offers spanning 10 product categories, with up to 200 product offers per category. Each category features between 8 and 15 attributes, resulting in a total of 115 unique attributes. Attributes with identical names across different product categories are treated as distinct.

AE-110K. The AE-110K (Xu et al. [4]) dataset comprises triples of product titles, attributes, and attribute values from the Sports & Entertainment category of the e-commerce platform *AliExpress*. Product offers are grouped by title, resulting in a subset of 10 categories, with up to 400 offers per category. Each category contains between 6 and 17 attributes, leading to a total of 101 unique attributes. The most frequent attributes in this dataset were *Brand Name*, *Material*, *Color* and *Category*.

Training/Test Split. Considering the sizes of the datasets, the AE-110K dataset is chosen for fine-tuning models due to its extensive number of examples, making it well-suited for training purposes. The dataset is divided into a 70:30 split, where 70% of the data is used for training and the remaining 30% is reserved for testing. On the other hand, the OA-Mine dataset, being smaller in size and entirely human-annotated, serves as an excellent benchmark for evaluating the models fine-tuned on the AE-110K dataset. This evaluation assesses the model’s generalization capability to perform on datasets containing different types of products and attributes.

3.2. Evaluation Metrics

For the task of attribute-value extraction from product titles, we adopt evaluation metrics used in previous work to assess our models’ performance. Specifically, we utilize the Exact Match (EM) metric for generated values, which requires the entire sequence of the generated value to exactly match the corresponding gold standard value. Since values may consist of multiple tokens and models can generate these tokens in different orders, we also calculate the average Bag-of-Words precision, recall, and F1 score as additional evaluation measures, denoted as P , R , and $F1$, respectively. Let N represent the dataset size, $V = \{v_1, v_2, \dots, v_N\}$ be the gold standard values, $G = \{g_1, g_2, \dots, g_N\}$ be the generated values, and $|v_i \cap g_i|$ denote the Bag-of-Words overlap between the gold standard value and the generated value for the i -th instance. The formulas for computing precision (P) and recall (R) are as follows:

$$P = \frac{1}{N} \sum_{i=1}^N \frac{|v_i \cap g_i|}{|g_i|}, \quad R = \frac{1}{N} \sum_{i=1}^N \frac{|v_i \cap g_i|}{|v_i|}$$

3.3. Implementation Details

We utilized a 4-bit quantized version of the LLaMa-3.1 8B model, loaded via **Unsloth**, for both the text infilling and answer generation tasks. To enhance computational efficiency, we employed **Quantized Low-Rank Adaptation (QLoRA)** for fine-tuning, with a rank of 8 and an alpha value of 16. Both tasks were trained for a full training epoch using 70% of the AE-110K dataset. The AdamW optimizer was used with a learning rate of 2×10^{-4} . For the text infilling approach, we leveraged Infilling by Language Modeling (ILM) as discussed in one of the previous works (Donahue et al. [5]). The LLaMa language model was also fine-tuned as answer-generation task using alpaca prompt format. Both models were subsequently evaluated on the remaining 30% of the AE-110K dataset and the OA-Mine dataset.

3.4. Baseline

We compare our models with their non-fine-tuned counterparts to ensure a fair comparison, given that prior methods, such as SuoTag and BiLSTM-CRF, differ significantly in architecture. The relatively lower performance of these earlier approaches compared to modern LLM-based methods can be attributed to the requirement for a certain level of general world understanding when extracting attribute values.

Additionally, span-based models were not considered in our evaluation, as they are neither efficient nor accurate for this task. Specifically, span-based models can only extract a single attribute value at a time for a given attribute from a product title, whereas methods using LLM’s allow all attributes to be processed simultaneously, significantly reducing computational overhead.

Exploring Results

The exploration results aim to evaluate two critical aspects of model performance: robustness and versatility.

- **Robustness:** The model’s ability to maintain consistent performance across diverse scenarios, including both frequent and rare cases. By testing the model across these extremes, we ensure that its performance is not biased towards any specific subset of data.
- **Versatility:** The model’s capacity to generalize effectively across different datasets and attribute types. The versatility evaluation ensures that the model is not overly specialized, making it suitable for real-world applications involving diverse data sources.

Robustness

We observe that between ILM training paradigm and Question Answering finetuning, both trained for one full epoch on 70% of the dataset, the F1 scores are nearly identical, with ILM showing a slight advantage. A possible explanation for this could be that during QA fine-tuning, the loss reduction might stem more from optimizing the output structure(i.e input we are providing the attributes and the output as are extracting in json which contains attributes from input) rather than the actual output predictions.

Even with minimal training and the use of a quantized model fine-tuned with QLoRA at rank 8 (adding only 20M additional parameters), the scores are quite competitive compared to those achieved by T5 and GPT-2 in the original paper. However, the original paper does not explicitly detail the fine-tuning methodology, leaving room for interpretation and potential variations in comparison

Attributes	Model	EM	P	R	F1
Brand Name	T5	0.94	0.95	0.95	0.95
	Llama-QA	0.90	0.91	0.91	0.91
	gpt-2	0.94	0.94	0.94	0.94
	Llama-ILM	0.94	0.94	0.94	0.94
Material	T5	0.84	0.88	0.87	0.88
	Llama-QA	0.78	0.82	0.83	0.83
	gpt-2	0.85	0.88	0.88	0.88
	Llama-ILM	0.84	0.88	0.88	0.88
Color	T5	0.76	0.80	0.81	0.80
	Llama-QA	0.71	0.75	0.74	0.75
	gpt-2	0.75	0.80	0.79	0.80
	Llama - ILM	0.73	0.78	0.78	0.78
Category	T5	0.74	0.81	0.80	0.80
	Llama-QA	0.46	0.62	0.59	0.60
	gpt-2	0.79	0.81	0.81	0.81
	Llama - ILM	0.59	0.66	0.65	0.66

Note 1: Llama-QA is question answering tuned.

Note 2: Llama-ILM is Infilling language modelling.

Table 1. Frequent attributes

In the case of rare attributes, both LLaMA-ILM and LLaMA-QA outperform T5 and GPT-2. This is expected, as performance on rare attributes heavily relies on the model’s general knowledge. Even during fine-tuning, fewer examples of rare attributes are encountered, making the model’s pre-trained general understanding crucial for success.

Attributes	Model	EM	P	R	F1
Frame Color	T5	0.74	0.74	0.74	0.74
	Llama-QA	0.82	0.84	0.85	0.84
	gpt-2	0.69	0.69	0.69	0.69
	Llama-ILM	0.89	0.89	0.89	0.89
Lens Color	T5	0.69	0.69	0.69	0.69
	Llama-QA	0.6	0.6	0.6	0.6
	gpt-2	0.67	0.67	0.67	0.67
	Llama-ILM	0.79	0.79	0.79	0.79
Shell Material	T5	0.58	0.68	0.77	0.72
	Llama-QA	0.83	0.87	0.91	0.89
	gpt-2	0.47	0.59	0.72	0.65
	Llama-ILM	0.55	0.61	0.66	0.64
Wheel Material	T5	0.78	0.78	0.78	0.78
	Llama-QA	0.83	0.83	0.83	0.83
	gpt-2	0.69	0.69	0.69	0.69
	Llama-ILM	0.78	0.83	0.89	0.86
Product Type	T5	0.52	0.62	0.64	0.63
	Llama-QA	0.62	0.65	0.66	0.65
	gpt-2	0.57	0.68	0.68	0.68
	Llama-ILM	0.91	0.92	0.92	0.92

Table 2. Rare attributes

Versatility

Consider the OA-Mine dataset, where the instruct model achieved a significantly higher F1 score compared to the fine-tuned model and ILM model. This highlights a potential issue with fine-tuning: it can degrade the model’s generalized performance.

On the other hand consider the AE.110k dataset, where fine-tuned models demonstrate a significantly higher F1 score compared to their instruct model counterparts. This suggests that fine-tuning on a specific dataset can substantially enhance performance for that dataset.

Dataset	Model	F1-score
OA-Mine	llama-8b	0.72
	ft-llama-8b-ILM	0.57
	ft-llama-8b-QA	0.62
AE.110k	llama-8b	0.60
	ft-llama-8b-ILM	0.81
	ft-llama-8b-QA	0.77

Table 3. finetuned v/s pretrained

Note 1: Llama-8b is instruction fine tuned.

Note 2: ft-Llama-8b is finetuned on the AE.110k dataset in 2 methods.

Conclusion

From this experiment, we conclude that, unlike the ExtractGPT paper where GPT-3.5 and GPT-4 were used for fine-tuning (which is resource-intensive), smaller LLMs show significant gains even with QLoRA, and the training process is simpler. Therefore, it is preferable to fine-tune models for individual datasets rather than using a single generalized model for smaller LLMs.

Additionally, ILM paradigm is a better choice compared to Question Answering fine-tuning, as it consumes fewer tokens resulting in lower compute requirements during both training and inference without sacrificing performance.

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